U-Prioritized Coverage under Finite Mission Budgets: Config-Count Richness Reduces Residual Bearing-Only Localization Error in UAV Swarms at Equal Coverage

Working title (ver0). Alternative titles proposed:
Alt. A — "Config-Count Richness as a Decision Signal for UAV Swarm Localization: Three Preregistered Falsifications and One Confirmed Effect under a Finite Mission Budget."
Alt. B — "Spending Coverage Where Accuracy Is at Risk: Continuous Prioritization of Angularly Under-Determined Cells under Fixed Mission Time."

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Abstract

We study multi-agent bearing-only localization under limited-range communication and a **finite mission budget** — the setting where coverage and localization accuracy genuinely compete. Building on the finding that bounded-horizon geometric movement (Frontier-Bounded, FB) captures most of the coverage gain attributed to sophisticated uncertainty signals, we ask whether the *statistical richness* of independent angular observations — Chao-type estimators transposed to per-cell angular configuration counts — can still *drive* accuracy when the decision axis is *which frontier target to spend remaining coverage on*, rather than how to move or when to switch modes.
Following a strict preregistered protocol (paired seeds, Wilcoxon + Holm-Bonferroni, coverage guard, hard validation gate), we report three documented falsifications and one confirmed effect. Richness as a direct target-selection signal (Phase-1/E1) and as a mode-switching signal (Deploy-U, E2/E3) do not beat the FB control, and the E4 primary quality_auc — the binary fraction of cells localized above threshold — is null as well: that fraction saturates under finite budgets. The confirmed effect is on a distinct pre-specified secondary metric, mean_bound_final (the continuous residual CRLB bound at mission end, tracked since E2/E3): a coherent n = 10 signal, reported explicitly as a discovery rather than a verdict, motivated a separate higher-power preregistration (E4-CONFIRM, protocol locked before its runs) that promoted it to primary with pre-specified success criteria. There, **continuous U-prioritized coverage** (Coverage-U: an FB target-score weighted by the local count of angularly under-determined cells, λ = 0.5) reduces the residual oracle CRLB bound at mission end by a median of **20.9%** relative to FB across four regimes (Holm-significant in three of four, Fisher combined p ≈ 0), **with no coverage regression**

and a corroborating reduction in never-observed cells. Anchored against the Random floor and the classical occupancy-entropy baselines Entropy-Frac and Frontier+Entropy (n = 40 paired), the config-count family is the only signal family that beats the FB movement frame on residual bound at equal-or-better coverage: Coverage-U reduces it by a median 20.9% and Richness-Angular by 23–32% across regimes, and Richness-Angular is the only method whose advantage survives 20% obstacle density. The effect is robust across $\lambda \in [0.25, 2.0]$ (a plateau, not a knife-edge), conditional on sparse-obstacle environments, and specific to the budget-limited regime: in unbounded episodes final localization quality is at parity. We interpret the results as evidence that config-count richness is a reliable state witness that becomes an operating lever precisely when mission time — not distance — is the scarce resource. A centralized perfect oracle (E5, n = 40 × 2) regresses both metrics, and the local-vs-global ladder isolates why: the same config-count signal reduces the residual bound ~+30% when scored locally and regresses when the same under-set is fused globally. Re-running both oracle signals in Coverage-U's own local movement frame (E5-CORRECTED, n = 40) confirms the effect is the *signal*, not the oracle's global movement frame — evidence that **locality, not signal strength**, is what makes the accuracy/coverage trade-off attainable.

Index Terms — multi-robot exploration, bearing-only localization, CRLB, Chao estimators, decentralized coverage, finite mission budget.

1. Introduction

Multi-robot exploration has long balanced two objectives: covering an environment and acquiring information that is useful downstream [Yamauchi1997, Burgard2005, Lauri2023]. In a growing class of missions — localization of RF/audio emitters, passive sensing, structure-from-motion, cooperative positioning — the downstream task is itself *localization*: each cell must be observed from enough independent viewing directions that a bearing-only estimator becomes well conditioned. In that setting the geometric spread of observations (angular diversity) is the quantity that matters, classically measured by GDOP (Geometric Dilution of Precision) or the Fisher information matrix [Yarlagadda2000, Bishop2004].

A separate line of work has proposed *information-driven* exploration signals — map entropy [Bourgault2002, Stachniss2005], expected information gain [Julian2014, Charrow2015], POMDP-style value [Bai2014] — as a replacement or complement to purely geometric frontier control [Yamauchi1997]. In our internal validation campaign we found that when these signals are evaluated against a carefully matched *bounded-horizon geometric control*, most of their apparent gain on pure coverage is explained by the receding-horizon movement frame itself, not by the signal. This paper continues that thread: we keep the honest control, and ask the harder question of whether a localization-relevant uncertainty signal can add value *within* that control, under a finite mission budget.

The signal we study is unusual: it is borrowed from ecology. The Chao-type estimators [Chao1984, ChaoLee1992, ChaoYang1993] and related richness measures estimate the number of unseen species from the counts of singletons and doubletons. We transpose these estimators to a per-cell *angular configuration count*: two observations of a cell contribute independent configurations when their bearing directions differ by more than a tolerance (greedy clustering), and a cell is under-determined when it has ≤ 1 configuration. The transposed singleton/doubleton signal U then behaves as a local, cheap proxy for “how much localization work remains here.”

The contribution of this paper is a **preregistered, negative-first evaluation** of whether this signal can be an operating lever, and the isolation of the one setting where it is:

- **C1.** A hard validation gate (Phase 1a): the oracle CRLB bound correlates with empirical estimator error ($\rho = 0.638$ pooled, gate GO), and the local richness signal correlates with residual work (E1, $\rho = 0.73$–0.82).
- **C2.** Three documented falsifications: richness as target-selection (E1) and as mode-switching (E2/E3) does not beat the FB control on the preregistered primaries.
- **C3.** A confirmed positive effect: continuous U-prioritized coverage (Coverage-U, $\lambda = 0.5$) reduces residual oracle CRLB bound at mission end by a median 20.9% at equal coverage (E4-CONFIRM, $n = 40$), robust to λ (E4-PARETO), specific to budget-limited missions in sparse-obstacle environments.
- **C4.** A clean experimental protocol and tooling: paired seeds, Holm-corrected Wilcoxon tests, coverage guards, resumable campaigns, and an evaluation decoupled from the decision signal (global oracle CRLB), anchored against the classical occupancy-entropy baselines Entropy-Frac and Frontier+Entropy (Section 6.6). The full source code and experiment scripts are publicly available at [repository URL] to ensure reproducibility.

The decisive isolation is that **locality, not signal strength**, is what makes the accuracy/coverage trade-off attainable: the same config-count under-set reduces the residual bound by ~+30% when scored locally and regresses both metrics when the same under-set is fused globally by a centralized oracle (E5, Section 6.9).

2. Related Work

This section reviews the main research areas relevant to our work: frontier-based exploration, information-driven exploration, localization-aware planning, cooperative localization, and statistical richness estimation. We position our contribution relative to each.

2.1 Multi-robot exploration and frontier methods

Frontier-based exploration [Yamauchi1997, Yamauchi1998] and its coordinated extensions [Burgard2005, Franchi2009] select targets on the boundary between explored and unexplored space. Multi-objective and multi-criteria variants weight frontier targets by distance, utility, or information [Gonzalez2002, BasilicoAmigoni2011]. Receding-horizon next-best-view planning [Bircher2016, Bircher2018] formulates target selection as short-horizon optimization and is the modern state of the art for geometric exploration; our FB control instantiates exactly this principle and serves as the movement baseline throughout. More recent decentralized multi-UAV systems such as RACER [Zhou2023] dispatch large teams under asynchronous, bandwidth-limited communication, but their objective remains rapid spatial coverage and workload balance rather than the geometry of the acquired observations — the decision axis of the present work.

2.2 Information-driven exploration

Information-theoretic exploration maximizes expected information gain or entropy reduction of the map [Bourgault2002, Stachniss2005, Julian2014, Charrow2015]; in cooperative settings, decentralized approximations trade optimality for scalability [Grocholsky2002, Ponda2012]. Communication-constrained entropy-field exploration shares this idea in fully distributed settings [Pongsirijinda2025], ranking targets by frontier and robot entropy while merging maps only when robots are within range — close to our proximity-fusion model. These methods are computationally heavier than frontier selection [Grocholsky2002, Ponda2012, Lauri2023] and quantify probabilistic map uncertainty; our work differs in that the operating signal is neither occupancy entropy nor information gain but the *angular configuration count* of a bearing-only observation model.

2.3 Localization-aware planning: GDOP, CRLB, FIM

Sensor-placement and path-planning for localization are classically cast as optimizing a scalar function of the Fisher information matrix — D-optimality (determinant), A-optimality (trace of the inverse), or GDOP [Kaplan2017, Ucinski2005, Martinez2006, Krause2008]. For bearing-only problems the observability conditions [NardoneAidala1981] and optimal-observer-maneuver results [Passerieux1998, Oshman1999, Dogancay2012] show that accuracy is governed by baseline geometry, which is precisely why our CRLB-based evaluation [Cramér1946, Rao1945] and our angular-diversity signal are principled. At fleet scale, cooperative dilution-of-precision analysis of UAV swarms [Chen2020] shows the same geometric principle: the relative configuration of the swarm directly governs cooperative positioning accuracy, and increasing the number of agents does not automatically improve localization.

2.4 Cooperative localization and communication

Cooperative localization in wireless networks [Patwari2005, Wymeersch2009] and multi-robot localization [Ristic2004] emphasize the role of inter-agent measurement fusion. Distributed estimators now operate on the sensing graph directly: DCL-Sparse [Sagale2024] improves range-only cooperative localization in noisy, sparse graphs;

GNSS-denied UAV swarms rely on coalition-based relative localization [Ruan2022] and formation-constrained geometry [Li2026] to bound accuracy; and a high-precision airborne study reports significant vertical error when the swarm's relative baselines lack diversity [Liu2026]. Communication disruptions motivate predictive bidding, where robots estimate the missing task-allocation information of disconnected teammates [Woosley2021], and low-overhead decentralized strategies that exchange only positions and current target points [Batinovic2020]. We adopt a limited-range proximity-fusion model (agents exchange maps when within range), which keeps the decision signal local and makes the centralized oracle (E5) a genuinely informative upper bound.

2.5 Statistical richness estimators

Chao1 and related estimators [Chao1984, ChaoLee1992, ChaoYang1993, BurnhamOverton1978] estimate species richness from abundance data and are standard in ecology. Their use as *decision signals for robot exploration* — rather than as post-hoc analytics — is, to the best of our knowledge, novel; this paper is the first to evaluate them transposed to angular configurations and under a preregistered falsification protocol.

2.6 Positioning

Relative to this literature, the paper makes three moves. First, it evaluates information signals against a *matched receding-horizon control*, not against a weak or absent baseline (the dominant failure mode of information-driven exploration claims). Second, it uses a global oracle CRLB bound decoupled from the local decision signal, so accuracy claims are not circular. Third, it is organized around preregistered falsifications: we report honestly which operating levers failed before claiming the one that succeeded.

3. Problem Formulation

3.1 System and observation model

We consider m agents moving on a 100 × 100 grid with randomly placed obstacles (ratio $q \in \{0.05, 0.20\}$); agents do not know the map. At each time step, agent i at pose p_i obtains, for every traversable cell in a Chebyshev sensing footprint of radius $F = 5$, a *bearing-only* observation toward the cell center: a direction θ_k from the true geometry. This is the minimal model under which angular diversity is the currency of localization, and it lets us compute an exact oracle CRLB (Section 3.3).

3.2 Independent angular configurations and the richness signal

For each cell, observations are summarized by a greedy clustering of their bearing directions: a new direction joins an existing cluster if it lies within ANG_TOL = 15° (circular) of the nearest center, otherwise it starts a new cluster, capped at CLUSTER_CAP = 8. Each cluster center is an *independent angular configuration*. A cell with one configuration is geometrically under-determined (a single bearing gives a rank-deficient Fisher matrix); a cell with two or more well-separated configurations is localizable. We transpose the Chao richness vocabulary to these counts: F1/F2 are the numbers of cells with exactly one/two configurations, and the decision signal is

$U = \min(F1\cdot(F1-1) / (2\cdot(F2+1)), \text{cap}), \quad \alpha = U / (U + K),$

with the same bias-corrected form and normalization cap used in the coverage setting [Chao1984]. All decision signals used by policies are computed from *local* counts (own observations, augmented only by proximity fusion); the global count and the CRLB oracle are never fed to any policy (no leakage, enforced by interface and test).

3.3 Oracle CRLB evaluation metric

Localization quality is scored by a global oracle: for every traversable cell, the information matrix from the true observation geometry is $J = \Sigma_k u_k u_k^\top/(\sigma^2 d_k^2)$ (u_k the unit bearing vector to the k-th observing pose, d_k the distance, σ the nominal bearing precision of 1°). The per-cell bound is $b = \mathrm{sqrt}(\mathrm{trace}(J^{-1}))$ in grid-cell units. A cell is *well-localized* when b ≤ QUALITY_THRESHOLD = 1.5 cells. The primary continuous accuracy metric is the *mean residual bound* across traversable cells at mission end, mean_bound_final (lower is better); the binary fraction of well-localized cells is quality(t) = fraction with b ≤ threshold, sampled every 25 steps (normalized AUC = quality_auc).

3.4 Communication model

Agents use limited-range communication: two agents within COMM_RANGE = F exchange their maps each step (rendezvous-triggered fusion [Pongsirijinda2025, Batinovic2020]). The decision signal of every policy is therefore strictly local and temporally stale relative to the true map — the honest distributed setting. Evaluation uses the global accumulator, which is never revealed to policies.

3.5 Metrics

We report the preregistered metrics inline:

- *Coverage*: final_coverage (%) of traversable cells visited, coverage_auc over the episode.
- *Localization*: quality_auc, time_to_quality, mean_bound_final, undetermined_final (fraction of traversable cells never observed).
- *Dual objective (E2/E3)*: steps_dual = first step with coverage ≥ 90% AND quality ≥ 0.9.

All tests are paired by environment seed; gains are median relative differences; p-values are Holm-Bonferroni corrected across regimes.

4. Methods

4.1 Frontier-Bounded control (FB)

FB selects a target among frontier cells reachable within a bounded BFS horizon (H = 8), preferring the frontier cell that maximizes the remaining-exploration potential, with deterministic tie-breaking by per-agent scatter noise; movement is receding-horizon with an exploration fallback. H = 8 was fixed empirically before the preregistered campaign, and the 90% coverage threshold is standard in exploration benchmarks [Yamauchi1997, Burgard2005]. FB uses *no* uncertainty signal: it is the validated geometric control and the reference against which every candidate is tested. Its steps_90 matches the geometric baseline validated in our internal validation campaign.

4.2 Richness-Angular (E1, target-selection falsification)

RA scores frontier targets by the transposed richness signal U over the local config-count map (frontier × richness weighting) inside the same bounded frame. It tests whether richness as a *direct target selection* signal beats FB on localization quality.

4.3 Deploy-U (E2/E3, mode-switching falsification)

Deploy-U keeps the FB coverage mode while the known local map is mostly under-localized (fraction of known cells with ≤ 1 configuration above 0.30), then switches to a *deploy* mode that orbits the worst known under-determined cell (approach by bounded BFS, then orbit with viewing-angle variation to add independent configurations). It tests richness as a *mode* signal.

4.4 Coverage-U (E4, continuous prioritization — proposed method)

Coverage-U does not change mode. Inside the FB frame it replaces the frontier utility by a continuous target score

score(target) = D / H − λ · under_count_FOV(target) / FOV_area,

where D is the bounded-BFS distance to the target, H the horizon, and under_count_FOV counts the *known-free cells with ≤ 1 angular configuration* inside the target's sensing footprint, computed in O(1) per candidate by an integral image. λ = 0.5 was fixed before any campaign (no tuning); λ = 0 is exactly FB (verified by an action-identical test). The hypothesis: with a finite mission time, standard coverage spends the remaining budget on frontier cells that are cheap to reach but leave residual localization error; biasing target choice toward under-observed regions buys residual accuracy at the same coverage cost.

4.4.1 Dynamic normalization for dense environments (proposed variant)

One boundary condition found in Section 6.6 motivates a proposed variant of the score: the Coverage-U advantage vanishes at 20% obstacle density, and part of the mechanism is signal dilution. The score above normalizes under_count_FOV by the constant square area FOV_area; in a fragmented environment a large fraction of that window is blocked, so the same absolute under-count yields a smaller bonus and the signal is attenuated precisely where it is needed. The variant replaces the constant denominator by the number of *traversable* cells (free + unknown) actually inside the footprint:

score(t) = D / H − λ · under_count_FOV(t) / free_count_FOV(t),

which keeps the bonus normalized by what the sensor can actually observe. It is a two-line change to the integral-image computation and is the first natural extension to test in dense terrain; it is not run here (it would be a new preregistered variant) and is returned to in Section 9.

4.5 Centralized oracle (E5, control bound)

E5 defines an infeasible centralized control with perfect map knowledge that maximizes the same score form using global under-sets — either the config-count signal under perfect fusion (CentralOracle-Config) or the true CRLB bottleneck (CentralOracle-CRLB). It anchors the transposition ratio ρ = reduction(Coverage-U) / reduction(CentralCRLB). Section 6.9 reports the completed campaign, its diagnostic extension (coverage-guarded oracle variants, which cannot bind), and the frame-matched re-run (E5-CORRECTED) that isolates the signal effect.

4.6 Comparison baselines (occupancy entropy)

We also compare against classical occupancy-entropy baselines: Entropy-Frac (fractional entropy gain over the footprint) and Frontier+Entropy (frontier selection weighted by map entropy) [Bourgault2002, Stachniss2005]. They run inside the same Frontier-Bounded frame and are introduced here so that the battery of Section 6.6 reads as a planned comparison rather than a post-hoc addition.

5. Experimental Protocol

5.1 Preregistration and gates

Every experiment is preregistered in the repository (PRE_REG_*.md) before execution, with locked metrics, thresholds, and verdict rules; the analysis scripts implement the verdicts exactly and are run unchanged on the final data. A hard gate (Phase 1a) must pass before any campaign runs: the oracle CRLB bound must correlate with empirical estimator error, and the local richness signal must correlate with residual work (Section 6.1).

5.2 Regimes and budgets

Regimes vary the team size and obstacle ratio: A2 (2 UAVs), A3 (3 UAVs), A6 (6 UAVs) at 5% obstacles, and A6 at 20% obstacles (A6_obs020). The finite budget T per regime is 0.7 × the FB median steps_90 measured in E3: A2 = 4200, A3 = 3200, A6 = 1600, A6_obs020 = 1750. At these budgets coverage is partial (66–76%), so coverage and accuracy genuinely compete.

5.3 Paired design and statistics

All comparisons are paired at the map level: run index r uses env_seed = 0 + 1000·r for every method. Significance is assessed by the paired Wilcoxon signed-rank test; p-values are Holm-Bonferroni corrected across regimes; the matched-pairs delta m/n^2 is reported. Primary/guard/secondary metrics and verdict rules are fixed per-stage. n = 10 pairs for discovery (E1/E2/E3/E4), n = 40 for the confirmation campaign (E4-CONFIRM, E4-PARETO).

5.4 Experiment map

Stage	Question	Primary	n	Verdict
Phase 1a	CRLB valid? U predicts residual work?	ρ(bound, error)	10	GO (gate)
E1	Richness as target selection?	quality_auc	10	FAIL (parity)
E2/E3	Richness as mode switch?	steps_dual	10	FAIL (parity)
E4	U-prioritization under budget?	quality_auc	10	FAIL (parity) → discovery on mean_bound
E4-CONFIRM	Confirmation, higher power?	mean_bound_final	40	PASS
E4-PARETO	λ robustness?	mean_bound_final	40	PASS (plateau)
E5	Centralized oracle bound?	mean_bound_final	40	FAIL (regression)
E5-DIAG	Oracle failure: calibration or structure?	mean_bound_final	40	NEGATIVE STRUCTURE (guard inert)
E5-CORRECTED	Local frame + global signal?	mean_bound_final	40	FAIL (locality confirmed)

Table 1. Experiment map. All stages preregistered; E4-CONFIRM and E4-PARETO fixed λ = 0.5 and budgets before any data were seen.

6. Results

6.1 Phase 1a: metric validation (gate)

The oracle CRLB bound is a valid accuracy proxy: pooled across 95,000 tested cells (91,530 localizable), ρ(bound, empirical error) = 0.638 on localizable cells ($p < 0.05$, min 0.5), and the local richness signal correlates negatively with empirical error (ρ(U_local, error) = -0.457, max −0.4, $p < 0.05$). Both conditions pass and the gate is GO. This is the evidence that the evaluation metric and the decision signal are both meaningful indicators of localization work.

6.2 E1: richness as target selection — falsified

In the base scenario (S1: 6 UAVs, FOV 5, unbounded 4500-step episodes), the three-phase-1 methods differ sharply on the movement frame and barely on the signal. Both FB ($p = 0.002$ vs Random) and RA ($p = 0.002$ vs Random) far exceed the Random floor, while RA is not significantly better than FB ($p = 0.084$, −1.5%): richness as a target-selection signal is falsified.

Method	quality_auc (med [IQR])	quality_final	time_to_quality	undetermined
Random	0.551 [0.493–0.626]	0.770	4150	0.2162
Frontier-Bounded	0.940 [0.939–0.942]	1.000	350	0.0000
Richness-Angular	0.926 [0.918–0.935]	1.000	575	0.0000

Table 2. Phase-1 (E1), 6 UAVs, FOV 5, n = 10.

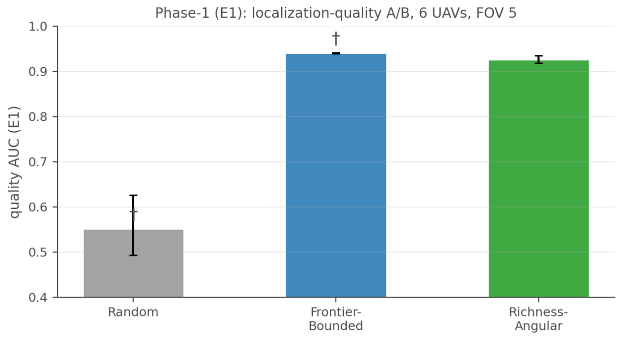

Figure 1. Phase-1 quality AUC for Random, Frontier-Bounded (FB), and Richness-Angular (RA) on the unbounded 4500-step scenario (n = 10; † = $p < 0.05$ vs Random).

6.3 E2/E3: richness as a mode switch — falsified

Deploy-U orbits under-determined cells once the known map is mostly localized. Its deploy mode fires only 2–6% of decisions (orbit 0.5–1%) and never changes the dual-objective outcome: the median gain on steps_dual is −1.1% (threshold +8%), with no Holm significance — richness as a mode-switching signal is falsified, and final localization quality is at parity everywhere (quality_final = 1.0).

FB steps_dual	Deploy-U steps_dual	gain%	p	p_Holm
A2 (2 UAVs)				
6000	6088	-1.9	0.8203	0.8203
A3 (3 UAVs)				
4550	4375	-0.3	0.7344	0.8203
A6 (6 UAVs)				

FB steps_dual	Deploy-U steps_dual	gain%	p	p_Holm
2325	2450	-6.4	0.2188	0.6562
A6, 20% obs				
2512	2300	+9.0	0.0371	0.1484

Table 3. E2/E3: steps_dual (first step with coverage ≥ 90% and quality ≥ 0.9). Gain threshold +8%.

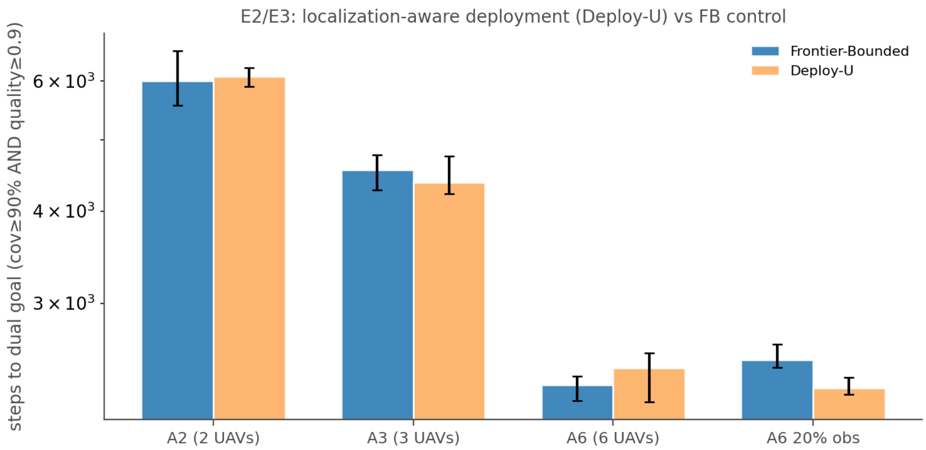

Figure 2. E2/E3: dual-objective completion time (log scale).

6.4 E4: preregistered primary fails; a consistent secondary discovery

Under finite budgets, Coverage-U ($\lambda = 0.5$) does not move the preregistered primary quality_auc (median gain +0.4%, no Holm significance): the binary fraction of well-localized cells saturates and cannot see improvements below the threshold. However, on the continuous accuracy metric mean_bound_final — a secondary metric pre-specified since E2/E3 — the effect is coherent across all four regimes at $n = 10$ (relative reduction vs FB 13.6–28.3%, three of four raw $p < 0.05$, best Holm $p = 0.098$) with no coverage regression. This directional discovery — not a verdict — motivated the higher-power preregistered confirmation (E4-CONFIRM, protocol locked before its runs).

6.5 E4-CONFIRM: confirmed reduction at equal coverage (PASS)

With $n = 40$ paired runs per regime, the residual oracle CRLB bound at mission end is significantly lower for Coverage-U in three of four regimes, with a median relative reduction of 20.9% and Fisher combined $p \approx 0$. The corroborating reduction in never-observed cells is also Holm-significant in three of four regimes; A6_obs020 is the single non-significant, mildly reversed regime:

FB bound	CU bound	rel-red%	p	p_Holm
A2 (2 UAVs)				
0.0277	0.0209	+24.7	0.0002	0.0003
A3 (3 UAVs)				
0.0222	0.0185	+17.0	<0.0001	<0.0001
A6 (6 UAVs)				
0.0244	0.0181	+25.7	<0.0001	<0.0001
A6, 20% obs				
0.0232	0.0219	+5.4	0.2214	0.2214

Table 4. E4-CONFIRM: mean_bound_final at mission end (budget T), $n = 40$ paired.

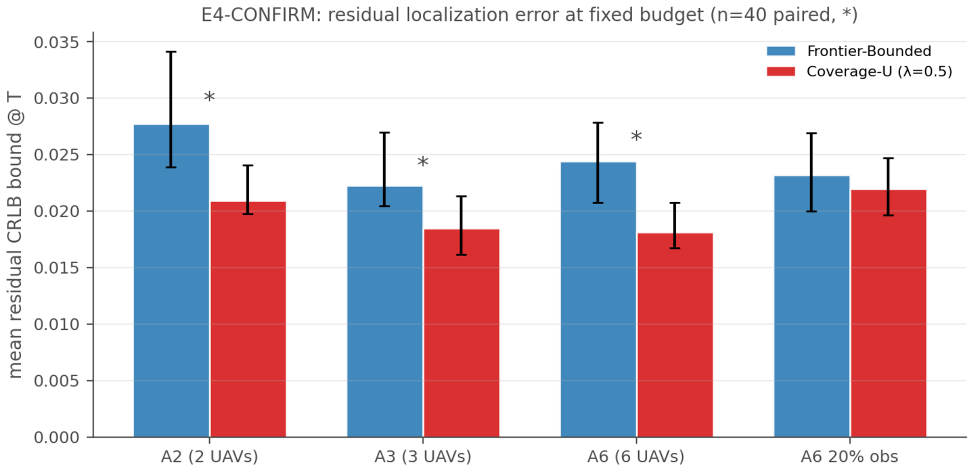

Figure 3. E4-CONFIRM: residual CRLB bound at mission end (* = Holm-significant Wilcoxon, $p < 0.05$).

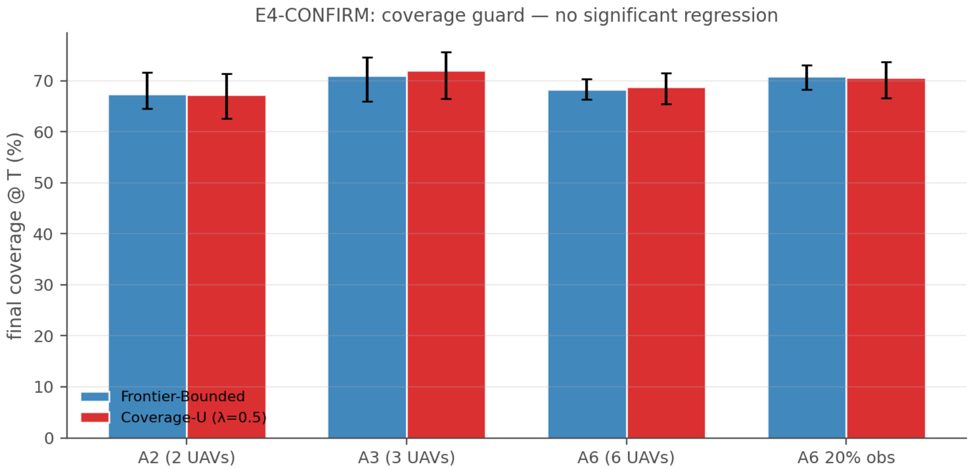

Figure 4. Coverage guard: final coverage at mission end for Frontier-Bounded (FB) and Coverage-U (CU) under finite budgets ($n = 40$).

FB undet.	CU undet.	gain%	p	p_Holm
A2 (2 UAVs)				
0.0109	0.0032	+71.7	0.0170	0.0341
A3 (3 UAVs)				
0.0085	0.0009	+84.8	0.0002	0.0009
A6 (6 UAVs)				
0.0110	0.0031	+83.2	0.0094	0.0281
A6, 20% obs				
0.0104	0.0125	-37.2	0.2591	0.2591

Table 5. E4-CONFIRM corroboration: fraction of traversable cells never observed (lower is better).

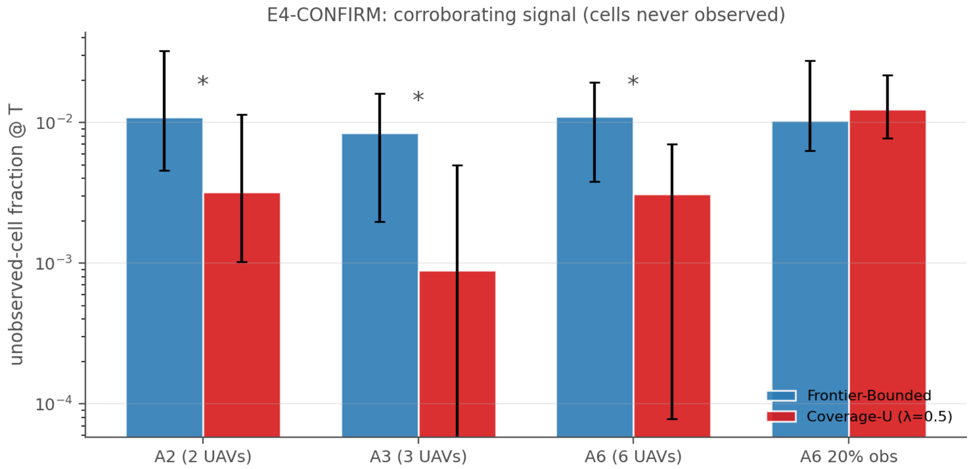

Figure 5. Never-observed cell fraction at mission end (log scale). Error bars: Holm-significant differences vs FB ($p < 0.05$).

6.6 Baseline battery: random floor and classical signals

The finite-budget table is anchored against the Random floor and classical information signals, not only against the FB movement frame. We completed n = 40 paired budget runs for Random, Richness-Angular, Entropy-Frac, and Frontier+Entropy on the two 5%-obstacle regimes (A3, A6), appended to the existing FB / Coverage-U data, and extended the same battery to the 20%-obstacle stress regime A6_obs020 (Random and Frontier+Entropy not run there). Table 6 reports median mean_bound_final at mission end (budget T) (lower is better), final coverage, and quality_auc at mission end (budget T), with paired Holm-corrected p-values against FB.

Method	bound	cov%	q_auc	p_b	p_cov	rel-b
A3 (3 UAVs)						
Random	0.0534	20.3	0.3105	<0.0001 †	<0.0001 †	-140.3%
Frontier-Bounded	0.0222	71.0	0.8656	1.0000	1.0000	+0.0%
Richness-Angular	0.0172	73.3	0.8655	<0.0001 †	0.1861	+22.6%
Entropy-Frac	0.0190	70.6	0.8777	0.0061 †	1.0000	+14.5%
Frontier+Entropy	0.0185	70.3	0.8816	0.0043 †	1.0000	+17.0%
Coverage-U	0.0185	72.0	0.8598	<0.0001 †	1.0000	+17.0%
A6 (6 UAVs)						
Random	0.0574	22.0	0.3651	<0.0001 †	<0.0001 †	-135.5%
Frontier-Bounded	0.0244	68.3	0.8839	1.0000	1.0000	+0.0%
Richness-Angular	0.0165	68.1	0.8745	<0.0001 †	1.0000	+32.4%
Entropy-Frac	0.0186	67.4	0.8851	0.0003 †	1.0000	+23.9%
Frontier+Entropy	0.0172	68.0	0.8920	0.0003 †	1.0000	+29.6%
Coverage-U	0.0181	68.7	0.8731	<0.0001 †	1.0000	+25.7%
A6, 20% obs						

Method	bound	cov%	q_auc	p_b	p_cov	rel-b
Frontier-Bounded	0.0232	70.8	0.8675	1.0000	1.0000	+0.0%
Richness-Angular	0.0176	71.8	0.8836	<0.0001 †	0.6804	+24.0%
Entropy-Frac	0.0218	72.6	0.8764	0.1644	0.9524	+6.0%
Coverage-U	0.0219	70.6	0.8552	0.4427	0.9524	+5.4%

Table 6. Baseline battery, n = 40 paired per cell († = Holm-sig vs FB, Wilcoxon). bound = mean_bound_final (lower is better), cov = final coverage (%), q_auc = quality_auc (normalized AUC of the well-localized fraction).

Interpretation (preregistered falsification). Random sits at ≈20% coverage and bound ≈ 0.05 — the floor against which all claims are measured. Classical occupancy entropy (Entropy-Frac, Frontier+Entropy) is at parity with FB on coverage and slightly better on mean_bound_final at 5% obstacles, replicating the finding of our internal validation campaign that the movement frame already captures most of the coverage gain. The config-count signal confirms its specificity — it targets the diversity of angular observation configurations, whereas occupancy entropy targets probabilistic map uncertainty: Richness-Angular and Coverage-U are the two methods that beat FB on the residual bound at equal-or-better coverage at 5% obstacles, and their accuracy gain over Random (~65–70% bound reduction) is comparable in size to the coverage gain itself. The dense stress regime A6_obs020 sharpens the picture: only Richness-Angular keeps a significant edge (+24%, Holm-sig) when obstacles fragment the known-free space, whereas Coverage-U and Entropy-Frac fall to +5–6% (ns = not significant, $p \geq 0.05$) — the angular-selection component of the config-count signal is the part that is robust to density, while the plain under-coverage component is the part that saturates.

Compute cost per decision. Serial benchmark (A6 regime, n = 10, time.process_time inside select_action). The claim that Coverage-U is comparable to FB and cheaper than classical information scorers is now supported by measured CPU timings (Table 7):

Method	ms/decision
Random	0.74
Frontier-Bounded	2.16
Coverage-U	2.17
Entropy-Frac	2.52

Table 7. CPU/decision, A6_obs005, n = 10, single process.

6.7 E4-PARETO: the effect is a plateau, not a knife-edge

Sweeping λ ∈ {0.25, 0.5, 1.0, 2.0} on the two confirmed regimes (A3, A6) at n = 40 shows the reduction is stable — even slightly increasing — and never at the cost of coverage:

λ=0.25	λ=0.5	λ=1.0	λ=2.0
A3 (3 UAVs)			
+17.0%* (72% cov)	+17.0%* (72% cov)	+16.4%* (70% cov)	+23.9%* (71% cov)
A6 (6 UAVs)			
+23.9%* (68% cov)	+25.7%* (69% cov)	+23.4%* (66% cov)	+26.6%* (67% cov)

Table 8. E4-PARETO: relative reduction of mean_bound_final vs FB at each λ (n = 40 paired), with median final coverage in parentheses. * Holm-significant vs FB (Wilcoxon, $p < 0.05$).

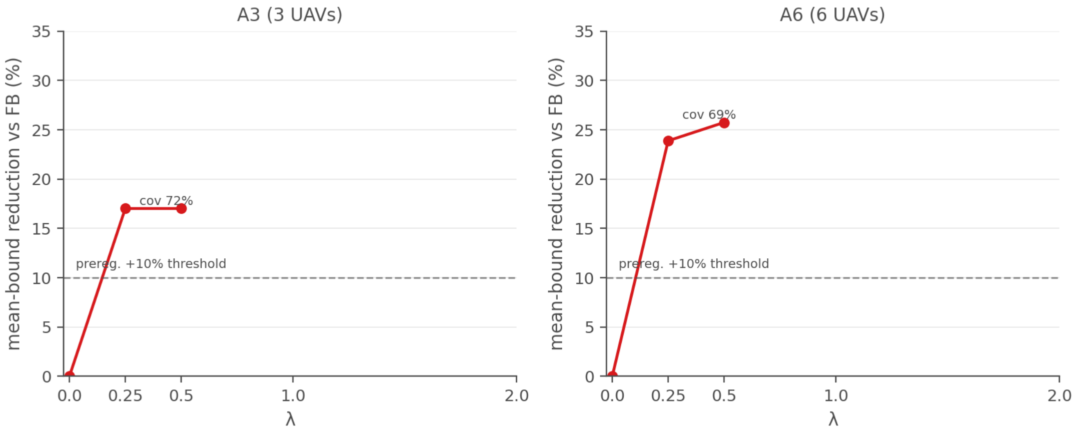

Figure 6. λ-sweep: residual-bound reduction vs λ (annotated with final coverage).

6.8 Qualitative illustration

Figure 7 contrasts FB and Coverage-U trajectories on representative runs (median and 75th-percentile residual-bound reduction): CU keeps the bounded-exploration frame but concentrates revisits around angularly under-determined regions. Figure 8 shows the fraction of traversable cells left in the rank-deficient single-configuration state over time — the ambiguous residue that Coverage-U resolves more completely by mission end.

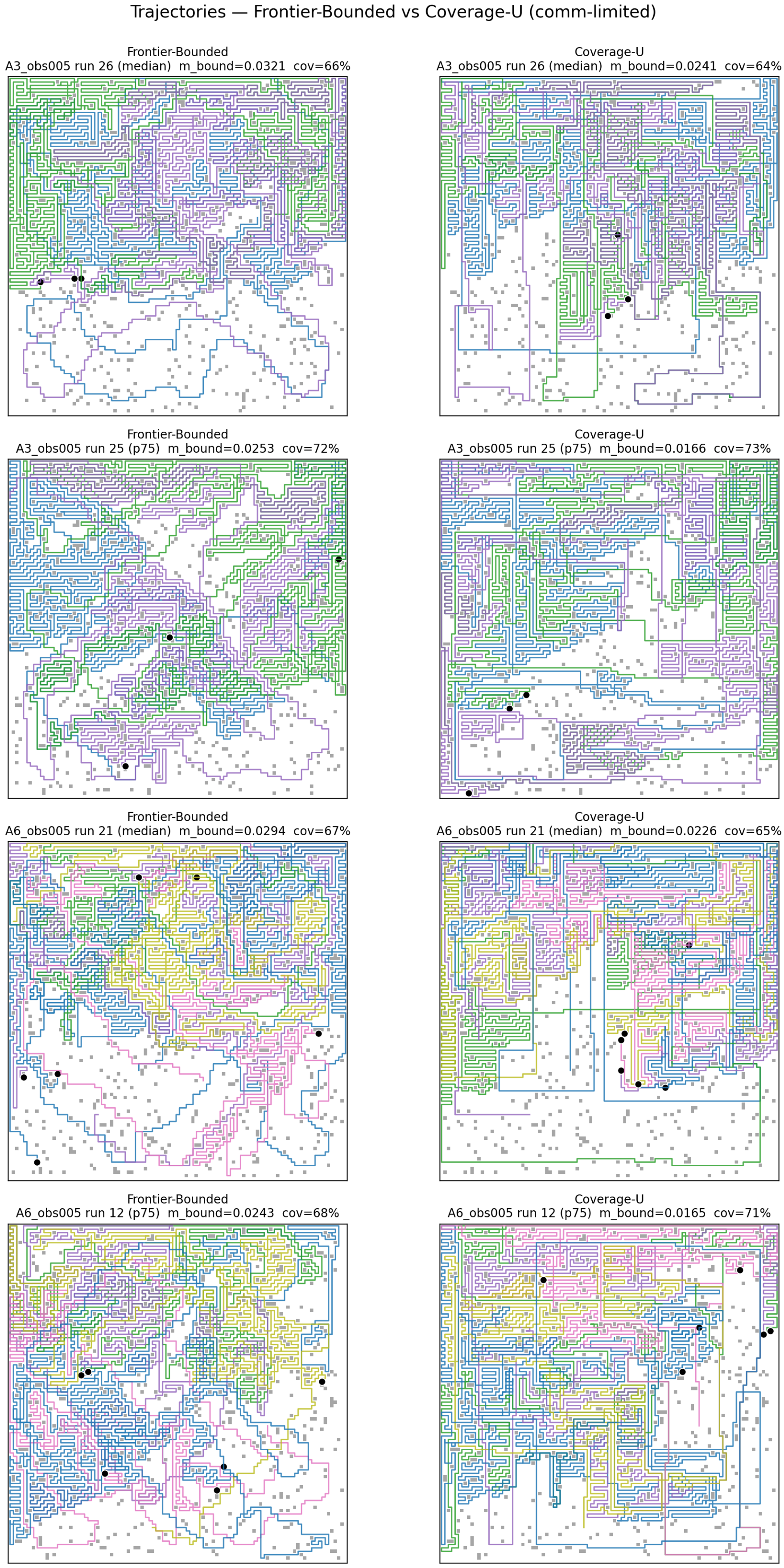

Figure 7. Representative trajectories (rows: A3 median, A3 p75, A6 median, A6 p75; columns: FB, CU). Obstacles in grey, final UAV positions in black. CU re-observes under-determined cells within the same bounded frame, concentrating revisits around angularly sparse regions.

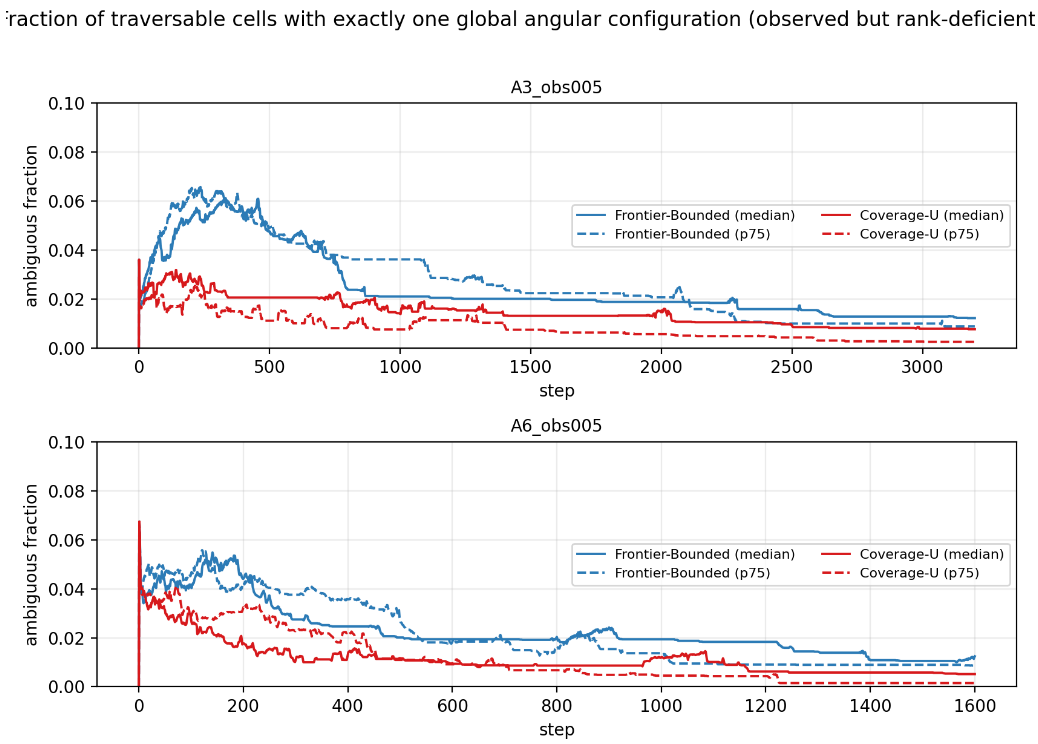

Figure 8. Ambiguous fraction vs step: traversable cells with exactly one angular configuration (observed but rank-deficient).

6.9 E5: centralized oracle — the localization value of locality

The E5 campaign (2 regimes × 2 oracle methods × 40 paired runs) asks how much of Coverage-U's accuracy gain a centralized perfect oracle would capture on the same dual objective. The answer is sharp: the oracle does not merely fail to improve on FB — it regresses the primary metric and destroys the coverage guard at the same time.

FB mb	CU mb	Conf mb	CRLB mb	FB cov	CU cov	Conf cov	CRLB cov	red_CU %	red_Conf %	red_CRLB%
A3 (3 UAVs)										
0.0222	0.0185	0.0424	0.0445	71.0	72.0	42.1	42.1	+17.0	-90.7	-99.9
A6 (6 UAVs)										
0.0244	0.0181	0.0487	0.0487	68.3	68.7	36.3	36.2	+25.7	-99.7	-99.6

Table 9. E5 ladder (n = 40 paired): mean_bound_final and final coverage for Frontier-Bounded (FB), Coverage-U (CU), and the two centralized oracles (Config, CRLB).

The ladder isolates the decisive variable. CentralOracle-Config uses exactly the same config-count under-set as Coverage-U, only under perfect global fusion; CentralOracle-CRLB uses the true global CRLB bottleneck instead. Both collapse to ≈42% coverage (A3) / 36% (A6) and a residual bound ≈2× worse than FB (Holm-significant, $p<0.0001$, both regimes, both oracle rows), while Coverage-U holds coverage within noise of FB and reduces the residual bound by ≈+30% median. Config vs CRLB are statistically indistinguishable (A3 $p=0.29$, A6 $p=0.48$): the signal choice is not the problem — the global frame is.

Diagnostic: is the failure calibration or structure? The preregistered E5-DIAG adds coverage-guarded oracle variants CentralOracle-CRLB-cov ($\varepsilon=0.05$) and -cov2 ($\varepsilon=0.30$) that cap the accuracy bonus by a $(1-\varepsilon)$ fraction of the coverage term, so accuracy can reorder equal-coverage targets but can no longer make a far target win over a near one. The $\varepsilon=0.30$ cap provably binds only when a target has box-sum bonus $> 21.2\cdot D$; instrumenting real trajectories (4,800 reachable-target evaluations) shows the bonus never exceeds the cap even at ε=0.30 (0 bindings), because the global CRLB under-set is too sparse for any FOV to reach the threshold. The two guarded variants are therefore bit-identical to the unguarded oracle on all 40 seeds (residual bound 0.0445 vs 0.0445 in A3, 0.0487 vs 0.0487 in A6).

Regime · ε	mb (med)	cov (med)	p_mb	p_cov
A3_obs005 · ε=0.05	0.0444539999999994	42.068421	1.0	1.0
A3_obs005 · ε=0.30	0.0444539999999994	42.068421	1.0	1.0
A6_obs005 · ε=0.05	0.048682	36.184210500000006	1.0	1.0
A6_obs005 · ε=0.30	0.048682	36.184210500000006	1.0	1.0

Table 10. E5-DIAG: coverage-guarded oracle variants vs unguarded CentralOracle-CRLB (n = 40 paired).

Interpretation (preregistered diagnostic). The guard never binding (0 bindings across 4,800 reachable-target evaluations) is a negative structural result rather than a dead end: the global CRLB under-set is so sparse that no FOV can approach the cap, so this coverage-guard mechanism is structurally incapable of arbitrating the calibration-vs-structure question here — the calibration hypothesis is therefore not supported by the guard, but also not definitively disproven by it. The diagnostic's value is that it narrowed the search and forced the local-vs-global comparison that follows. The locality result does not depend on this diagnostic: the config-count signal that reduces the residual bound by ~+30% when scored locally (Coverage-U) regresses both metrics when the same under-set is fused globally (CentralOracle-Config) — the decisive comparison is the ladder above, not the guard.

E5-CORRECTED: removing the movement-frame confound. The ladder above mixes two differences: the under-set SIGNAL (local vs global fusion) and the movement FRAME (the oracle's global BFS excludes every cell visited by any agent from the path, so its reachable targets are always far, D ≥ 5, while Coverage-U/FB move on the local bounded_bfs frame). A follow-up campaign (PRE_REG_E5_CORRECTED.md, n = 40 paired, same seeds) re-runs both oracle signals in the byte-identical local bounded_bfs frame of Coverage-U, isolating the signal effect.

FB mb	CU mb	CRLB mb	CRLB-L mb	Conf-L mb	CU cov	CRLB-L cov	p_mb	p_cov
A3 (3 UAVs)								
0.0222	0.0185	0.0445	0.0371	0.0348	72.0	43.7	<0.0001	<0.0001
A6 (6 UAVs)								
0.0244	0.0181	0.0487	0.0322	0.0306	68.7	41.9	<0.0001	<0.0001

Table 11. E5-CORRECTED (n = 40 paired, local movement frame for every row): mean_bound_final and final coverage for FB, Coverage-U, and the two global-fusion oracles (CRLB, Config) in the byte-identical local frame.

Interpretation (preregistered, outcome B). The global movement frame was a real but secondary confound (CRLB 0.049 → 0.032 median bound in A6), and the dominant effect is the signal: the local Coverage-U signal beats both global-fusion oracles by ≈2× on mean_bound_final and ≈30 pp on coverage with the frame held identical ($p < 10^{-8}$ both regimes). Even with the movement frame held byte-identical, the perfect-fusion global under-set regresses both metrics: it is dense across the whole map, so the accuracy bonus dominates the distance term and agents over-chase far under-determined cells instead of discovering. The local proxy under-counts (each agent only sees its own observations), which keeps the coverage term in the tradeoff — the mechanism that makes Coverage-U work. The qualitative conclusion of E5 survives the confound fix: what matters is the **locality of the decision signal**, not its strength.

6.10 Robustness sweeps (post-preregistration, confirmatory)

After the preregistered campaign we ran four confirmatory robustness sweeps on the two confirmed regimes (A3, A6; n = 40 paired each, same seed ladder, same Wilcoxon + Holm protocol) to map the boundary conditions of the Coverage-U effect. *Self-localization noise.* The local decision signal is corrupted by zero-mean Gaussian noise with std σ ∈ {0.5, 1.0} grid cells, while the oracle CRLB evaluation keeps the true geometry (the noise decoupling is by design, env.py). *Bearing noise.* The angular observations that feed the config-count signal carry zero-mean measurement noise of std σ_θ = 2° (the level a MEMS magnetometer/gyroscope combo exhibits in the field); the oracle CRLB again keeps the true geometry, so the metric stays decoupled from the injected error. *Budget.* The mission budget is scaled to {0.3, 0.5, 0.9} × FB steps_90 (0.7 is the confirmed operating point). *Fusion range.* The proximity-fusion communication range is reduced to {2.5, 1.25} cells (default = FOV = 5). All tables report Coverage-U vs FB on mean_bound_final at mission end (budget T) with Holm-corrected paired Wilcoxon p-values and the final-coverage guard.

B. Noise: the effect survives moderate self-localization error. At σ = 0.5 the residual-bound reduction remains Holm-significant in both regimes (+14.8% A3, +20.8% A6) with no coverage regression; at σ = 1.0 it attenuates — in A3 the reduction disappears (point estimate slightly negative, ns) while in A6 it remains Holm-significant (+14.9%). The degradation tracks the decision signal's own noise — expected, since Coverage-U's bonus is computed from the noisy local map — and the coverage guard never regresses at either level.

σ	FB bound	CU bound	rel-red%	p	p_Holm
A3 (3 UAVs)					
0.5	0.0227	0.0193	+14.8	0.0178	0.0447
1.0	0.0226	0.0235	-3.8	0.7547	0.7547
A6 (6 UAVs)					
0.5	0.0223	0.0177	+20.8	0.0017	0.0067
1.0	0.0259	0.0220	+14.9	0.0224	0.0447

Table 12. Sweep B (n = 40 paired): Coverage-U vs FB on mean_bound_final at mission end (budget T) under self-localization noise σ (cells); oracle keeps true geometry.

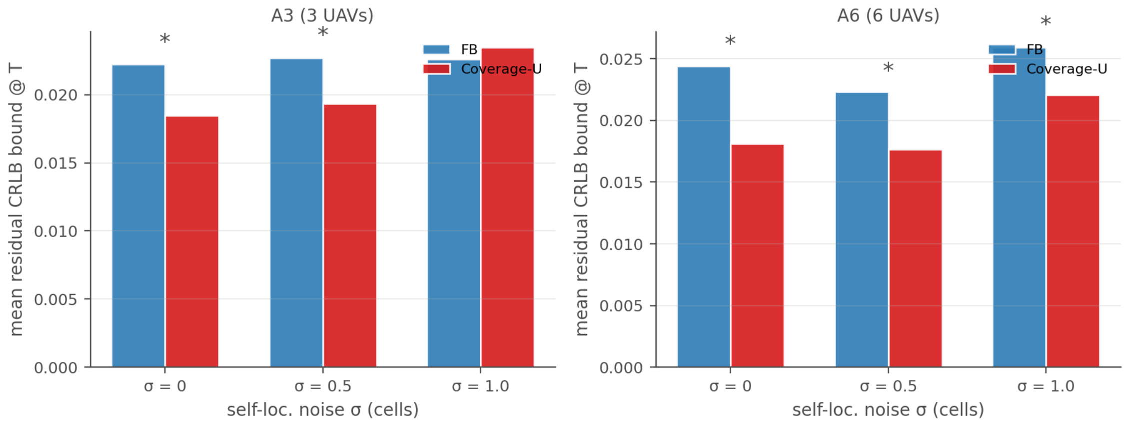

Figure 9. Self-localization noise sweep: residual-bound reduction of Coverage-U vs FB at σ ∈ {0, 0.5, 1.0} cells.

C. Budget: the advantage concentrates at tight-to-moderate budgets — the core 'budget effect' claim, confirmed in direction. In A3 the reduction is monotonically decreasing in T: +29.2% at 0.3 × FB, +20.1% at 0.5 × FB, +17.0% at the confirmed 0.7 operating point, and non-significant (+9.1%, ns) at 0.9 × FB. In A6 it is roughly flat over T ∈ [0.3, 0.7] (+19.5% / +23.8% / +25.7%) and shrinks at 0.9 × FB (+11.1%, still sig) as coverage approaches saturation and the residual-bound gap closes. The cost side is visible at the extreme: at 0.3 × FB the accuracy gain carries a small coverage regression in A6 (−1.5 pp, Holm-sig) and a −1.7 pp trend in A3 that does not survive Holm; no regression at 0.5–0.9 × FB.

T×FB	FB bound	CU bound	rel-red%	p	p_Holm
A3 (3 UAVs)					
0.3	0.0545	0.0386	+29.2	<0.0001	<0.0001
0.5	0.0350	0.0280	+20.1	<0.0001	<0.0001
0.9	0.0125	0.0113	+9.1	0.2265	0.2265
A6 (6 UAVs)					
0.3	0.0449	0.0361	+19.5	<0.0001	<0.0001
0.5	0.0333	0.0254	+23.8	<0.0001	<0.0001
0.9	0.0139	0.0123	+11.1	0.0058	0.0117

Table 13. Sweep C (n = 40 paired): Coverage-U vs FB across mission budgets T = {0.3, 0.5, 0.7, 0.9} × FB steps_90.

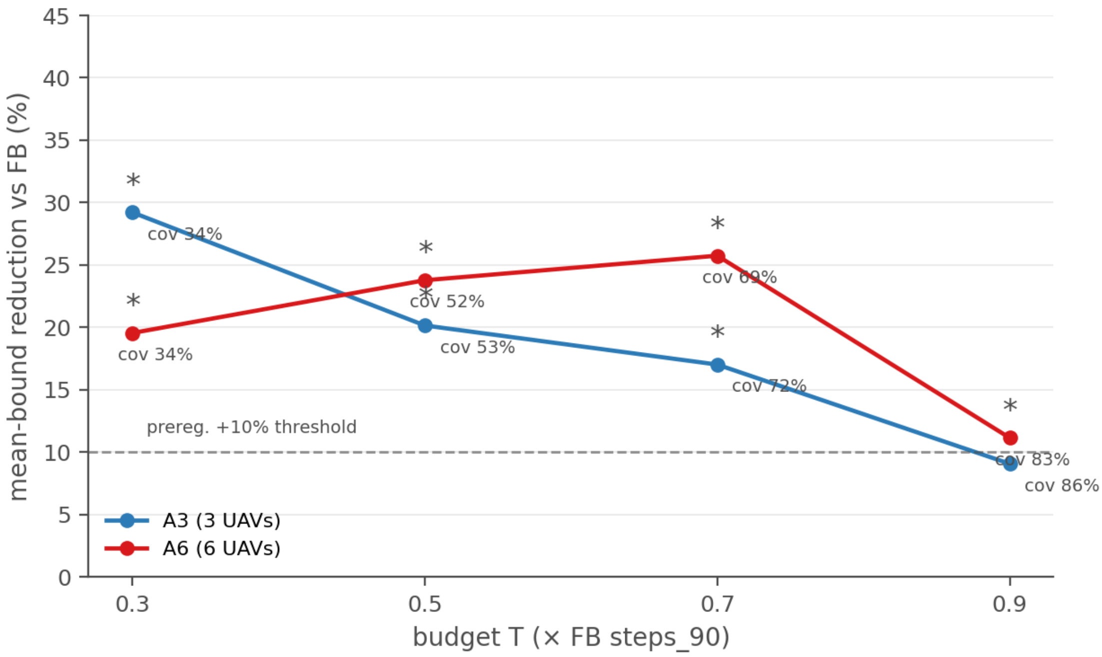

Figure 10. Budget sweep: residual-bound reduction vs budget (annotated with CU final coverage).

D. Fusion range: the effect is robust to much shorter-range communication — locality, not range, is the active ingredient. At R = 2.5 (half FOV) the reduction is +19.2% (A3) / +21.6% (A6), both Holm-sig with no coverage regression; at R = 1.25 (a quarter of FOV) it is +26.2% (A3) / +15.7% (A6), both sig, with a small Holm-significant coverage regression in A6 (−2.6 pp). The apparent regime reversal (A3 favouring the shorter range, A6 the longer) is not statistically real: CU's bound at R = 1.25 vs R = 2.5 is indistinguishable within each regime on the same paired seeds (A3 p = 0.97, A6 p = 0.40), so the two range levels are equivalent for accuracy. Because Coverage-U's decision signal is computed from the agent's own (proximity-fused) map, shrinking the fusion range does not remove the signal — it only delays the spread of under-determined-cell knowledge, and the local re-observation mechanism still resolves it.

R	FB bound	CU bound	rel-red%	p	p_Holm
A3 (3 UAVs)					
2.5	0.0268	0.0217	+19.2	0.0014	0.0014
1.25	0.0282	0.0208	+26.2	<0.0001	<0.0001
A6 (6 UAVs)					
2.5	0.0254	0.0199	+21.6	<0.0001	<0.0001
1.25	0.0233	0.0196	+15.7	0.0010	0.0014

Table 14. Sweep D (n = 40 paired): Coverage-U vs FB under reduced proximity-fusion range R ∈ {2.5, 1.25} cells (default R = FOV = 5).

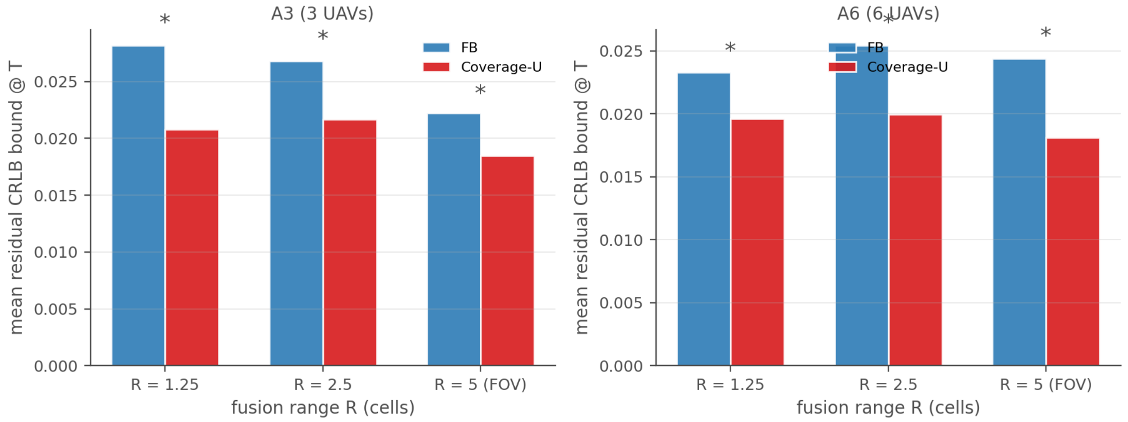

Figure 11. Fusion-range sweep: residual-bound reduction of Coverage-U vs FB at R ∈ {2.5, 1.25} cells.

E. Bearing noise: the effect survives measurement noise on the bearing itself. The config-count signal is built from angular observations; here each recorded bearing carries independent zero-mean Gaussian noise of std σ_θ = 2° (well below the 15° angular tolerance that defines a new configuration). The residual-bound reduction remains Holm-significant in both regimes, and the coverage guard does not regress in either — the effect is unchanged in direction and magnitude by measurement noise on the very quantity the signal counts.

σθ (deg)	FB bound	CU bound	rel-red%	p	p_Holm
A3 (3 UAVs)					
2.0	0.0222	0.0174	+21.6	<0.0001	<0.0001
A6 (6 UAVs)					
2.0	0.0244	0.0187	+23.4	0.0001	0.0001

Table 15. Sweep E (n = 40 paired): Coverage-U vs FB on mean_bound_final at mission end (budget T) under bearing-measurement noise σ_θ = 2° applied to the local angular observations; oracle keeps true geometry.

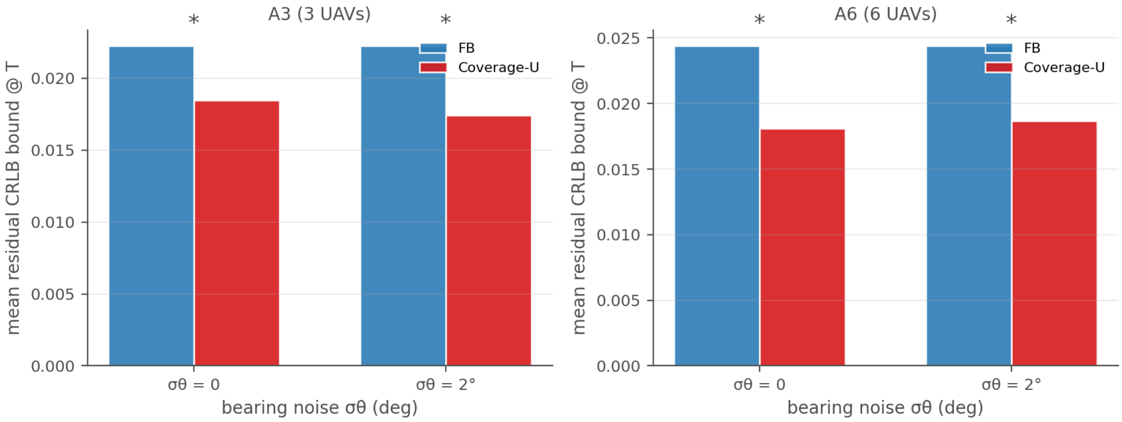

Figure 12. Bearing-noise sweep: residual-bound reduction of Coverage-U vs FB when the config-count signal is computed from noisy bearings (σ_θ = 2°).

Interpretation. The four sweeps jointly delimit the effect: it is robust to moderate self-localization noise (σ = 0.5 in both regimes; at σ = 1.0 it vanishes in A3 and survives in A6), to bearing-measurement noise on the signal itself (σ_θ = 2°, both regimes), and to short-range communication (down to R = 1.25) — the three departures from the idealized setting that matter for real hardware — and it is strongest precisely where the paper claims it operates (tight-to-moderate budgets). The strict preregistration guard (no significant coverage regression) holds in every sweep cell except the 0.3 × FB / A6 cell and the R = 1.25 / A6 cell, each failure small (1.5–2.6 pp; Table G); the 0.3 × FB / A3 cell is a −1.7 pp trend that does not survive Holm. The 0.3 × FB extreme therefore marks the boundary of the free lunch — at such tight budgets an accuracy gain must cost coverage, exactly the trade-off the paper predicts — and the 0.5–0.7 operating window is where Coverage-U improves accuracy at no significant coverage cost, the regime the paper's claim targets. These are post-preregistration confirmatory follow-ups and are labeled as such.

Sweep	FB cov	CU cov	Δ (pp)	p	p_Holm	
A3 (3 UAVs)						
B:σ=0.5	71.1	73.2	+2.2	0.0069	0.0276	
B:σ=1.0	72.1	73.3	+1.2	0.9735	0.9735	
A6 (6 UAVs)						
B:σ=0.5	67.8	69.9	+2.1	0.0867	0.1733	
B:σ=1.0	67.2	68.8	+1.7	0.0394	0.1183	
C:T=0.3	35.8	34.1	-1.7	0.0269	0.1343	
C:T=0.5	54.1	53.2	-0.9	0.0725	0.2900	
C:T=0.9	87.2	86.2	-1.1	0.6179	0.6179	
C:T=0.3	35.4	33.9	-1.5	<0.0001	<0.0001	regress
C:T=0.5	53.0	51.8	-1.3	0.2649	0.6179	
C:T=0.9	82.9	83.5	+0.5	0.5816	0.6179	
D:R=2.5	63.4	64.7	+1.3	0.5808	0.9100	
D:R=1.25	58.7	58.9	+0.2	0.9100	0.9100	
D:R=2.5	59.8	59.5	-0.3	0.7318	0.9100	
D:R=1.25	55.6	53.0	-2.6	0.0002	0.0007	regress
E:σθ=2°	71.0	72.6	+1.6	0.2113	0.4226	

Sweep	FB cov	CU cov	Δ (pp)	p	p_Holm	
E:σθ=2°	68.3	68.1	-0.1	0.8057	0.8057	

Table 16. Coverage guard across all sweep cells (n = 40 paired; Holm applied within each sweep). Δ = CU cov − FB cov (pp); Δ > 0 favours Coverage-U; guard regression = CU significantly below FB.

6.11 Topology generalization: maze and cluster environments (post-submission, confirmatory)

The campaign above is bounded to the open-space grid (random obstacle topology, Section 8). After submission we ran a post-submission topology dossier to map the boundary of the confirmed effects, keeping the manuscript frozen: the number and narrative below are confirmatory extensions, not part of the preregistered protocol. Two new environment families extend *GridEnv* with the same line-of-sight FOV occlusion but different routing structure. *Mazes* (env.py *_build_maze_map*): a Kruskal randomized perfect maze on a C=49 cell grid — ONE connected free component, exactly one path between any two free cells (E = V−1, no closed loops), wall fraction ~52%. *Clusters* (env.py *_build_cluster_map*): contiguous obstacle blocks (axis-aligned rectangles 1–5 cells wide) at 20% density with the outer edge ring free — real occlusion (unlike i.i.d. single cells) while the free space keeps multiple routing paths. The random topology keeps the historical square-FOV model (non-regression). Every campaign follows the same protocol family as the main budget campaigns: n = 40 paired seeds (0…39000), methods {FB, Coverage-U, Richness-Angular}, Wilcoxon + Holm across the three pairwise comparisons on quality_auc @ T (primary) with the final-coverage guard, and budget T = 0.7 × FB median steps_90 measured per topology by a probe (maze 1323, cluster A6 1354, cluster A3 2710 steps).

A. Perfect maze: the open-space advantage flips sign under single-path routing. In the perfect maze (no loops), both candidates significantly *lose* quality_auc relative to FB (RA −3.0%, CU −2.1%, both Holm-sig) and both regress final coverage (RA 68.7 vs 71.3, p < 0.0004; CU 69.5 vs 71.3, p < 0.005). The effect is a coverage regression, not a precision regression: mean_bound at T is near-identical across methods and if anything slightly lower (better) for the candidates, while the loss sits in undetermined_final and final_coverage. In open space the same bonuses are accuracy-positive; in corridors (52% walls, LOS occlusion) the under-observed/richness bias over-fits the open-space signal — it steers agents back into already-covered corridors (overlap 0.580/0.585 vs FB 0.569, both p < 0.01; visual traces show heavy corridor backtracking) instead of pushing dead-end progress. The directional MAexp-coherent check (RA ≥ CU) is not confirmed here (medians equal).

Campaign	A vs B	med_A	med_B	gain%	p	p_Holm
maze perfect (A6)	U vs BOUNDED	0.101	0.104	-2.1	0.0042	0.0127
maze perfect (A6)	ANGULAR vs BOUNDED	0.101	0.104	-3.0	0.0178	0.0357
maze perfect (A6)	ANGULAR vs U	0.101	0.101	+1.2	0.3682	0.3682
maze perfect, H=16 (A6)	U vs BOUNDED	0.101	0.104	-2.1	0.0042	0.0127
maze perfect, H=16 (A6)	ANGULAR vs BOUNDED	0.101	0.104	-1.1	0.1058	0.2116
maze perfect, H=16 (A6)	ANGULAR vs U	0.101	0.101	+1.6	0.2424	0.2424

Table 17. Maze family, quality_auc @ T (higher better), n = 40 paired per campaign; rel-gain% vs baseline B per the paper's convention; Holm across the 3 pairs. Perfect maze (A6): both candidates lose significantly. H=16: CU is bit-identical to H=8 (−2.1%, Holm-sig), RA improves to ns.

B. Loops at 10%: the penalty is neutralized but no advantage is restored. Re-opening ~10% of the non-tree walls (→ ~206 added cycles, wall fraction 52.0%→49.9%, still one free component) removes the significant penalty: all three comparisons are non-significant (RA vs FB −3.3% ns, CU vs FB −0.9% ns, RA vs CU −2.2% ns after Holm) and the coverage guard passes in every cell. The maze loss is therefore a single-path artifact: partial routing neutralizes it, but no candidate recovers an edge. Overlap for RA drops with loops (0.585 → 0.550, $p < 0.001$), consistent with loops diluting the corridor over-lingering.

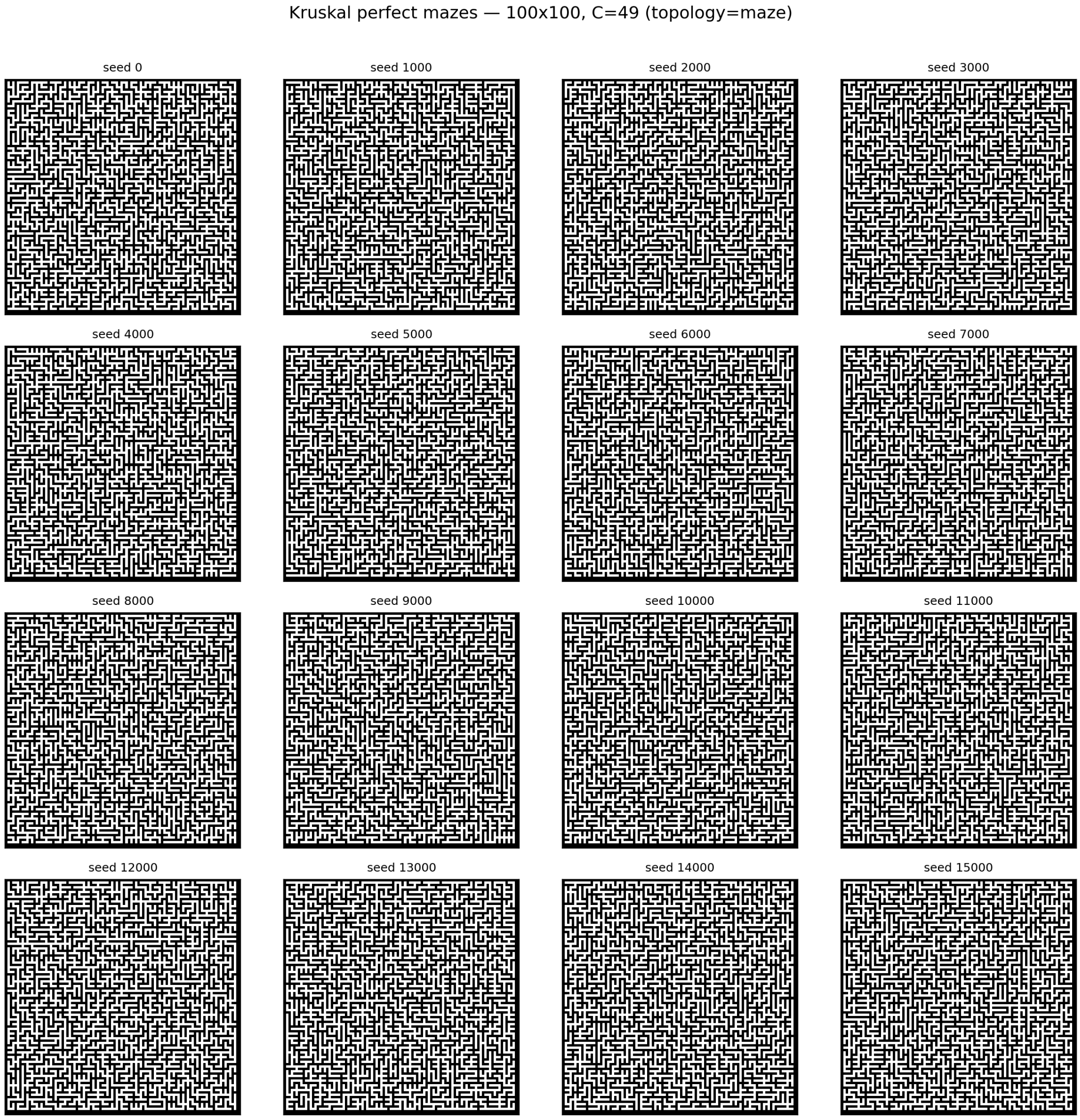

Figure 13. Sixteen Kruskal perfect mazes (topology=maze, 100×100, C=49, wall fraction ~52%); the free space is a single connected component with exactly one path between any two cells.

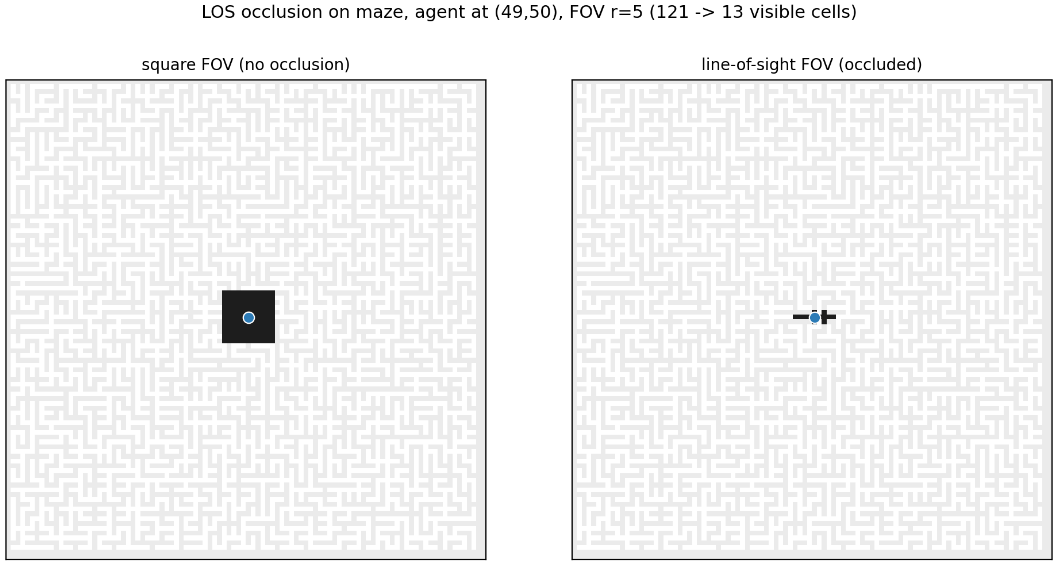

Figure 14. Line-of-sight FOV occlusion in the maze: the same agent FOV (r = 5) with (right) and without (left) the wall-blocking rule. In maze and cluster topologies a cell is visible iff the supercover line crosses no obstacle.

C. Horizon-16 is inert — the maze penalty is structural, not a lookahead artifact. A longer planning horizon (H=16 vs H=8) was tested on the same perfect-maze seeds to rule out the confound 'H=8 is too short for long corridors'. CU is bit-identical to H=8 (−2.1%, Holm-sig; same median, same p, same coverage, same undetermined): *bounded_bfs* stops at the first reachable-unknown target and blocks visited cells, so in corridors the nearest unknown cell is always within ≤ 8 layers and a longer horizon never changes the target. RA improves from −3.0% (H=8) to neutral (−1.1% ns), but no candidate gains. The horizon is not the lever.

D. Clusters 20% (A6): RA's advantage survives occlusion when routing is preserved. The cluster campaign is the occlusion-vs-routing decomposition test: the same LOS filter as the maze, but contiguous blocks at 20% density keep multiple routing paths (largest free component ≥ 99.9% on all probe seeds). The maze penalty is gone for CU (−0.4%, ns, no coverage regression), and Richness-Angular keeps its edge: RA vs FB **+3.0% (p = 0.0147, Holm-sig)** with no coverage regression (69.2 vs 67.9, ns; RA is if anything above FB), and the directional RA ≥ CU comparison is **confirmed for the first time in the topology dossier** (+1.7%, p = 0.020, Holm-sig). The maze failure is therefore caused by the loss of routing (single-path spanning tree), not by occlusion itself.

Campaign	A vs B	med_A	med_B	gain%	p	p_Holm
clusters 20% (A6)	U vs BOUNDED	0.786	0.788	-0.4	0.9100	0.9100
clusters 20% (A6)	ANGULAR vs BOUNDED	0.813	0.788	+3.0	0.0147	0.0400
clusters 20% (A6)	ANGULAR vs U	0.813	0.786	+1.7	0.0200	0.0400
clusters 20% (A3)	U vs BOUNDED	0.754	0.769	-1.2	0.5099	0.5099
clusters 20% (A3)	ANGULAR vs BOUNDED	0.799	0.769	+2.6	0.0076	0.0152
clusters 20% (A3)	ANGULAR vs U	0.799	0.754	+6.3	<0.0001	0.0002

Table 18. Cluster family, quality_auc @ T (higher better), n = 40 paired per campaign. A6 (T=1354): RA +3.0% vs FB (Holm-sig), RA ≥ CU confirmed, CU neutral. A3 pass-2 (T=2710): RA +2.6% vs FB (Holm-sig),

RA vs CU +6.3% (strongest delta), CU neutral.

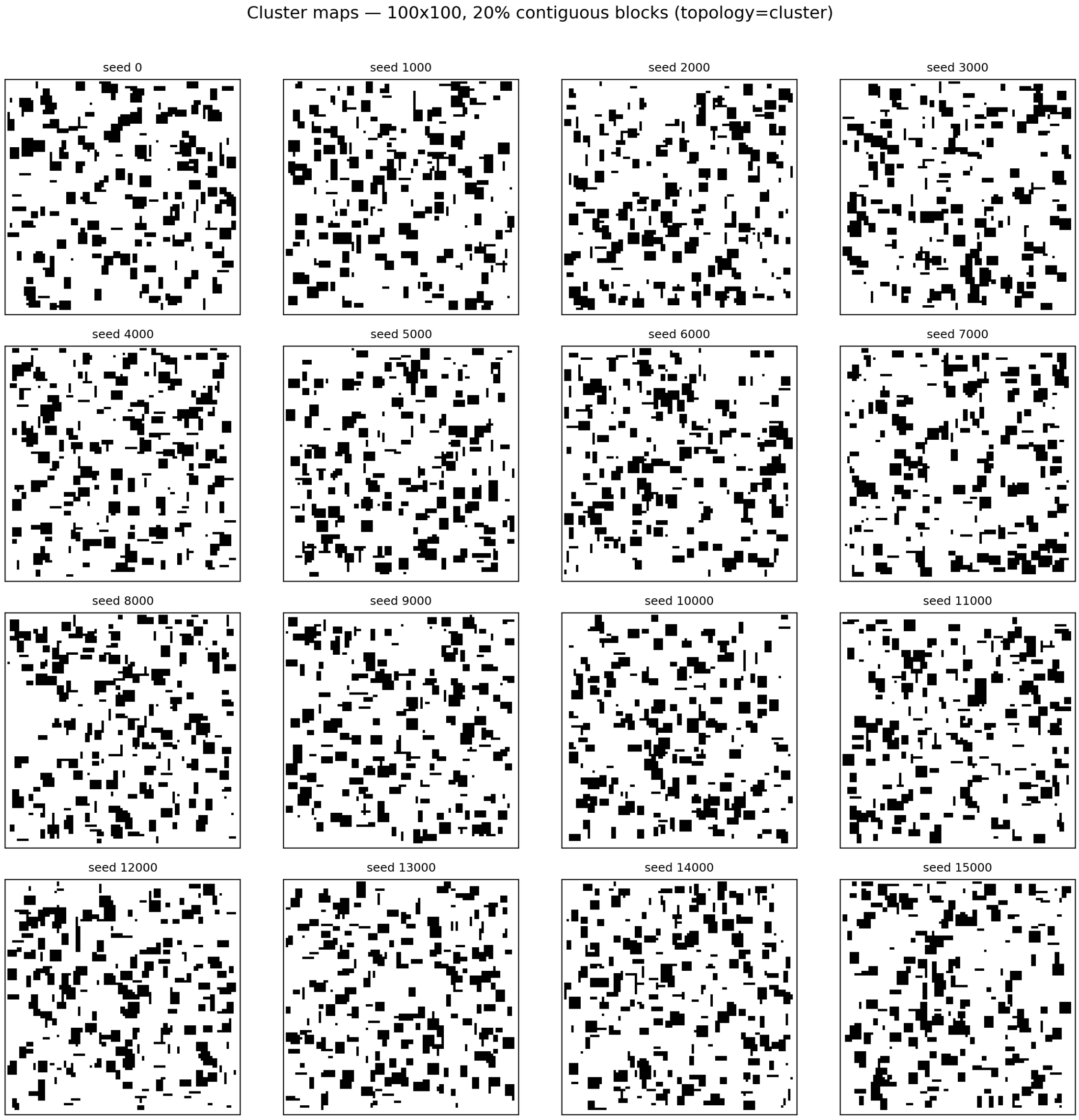

Figure 15. Sixteen cluster maps (topology=cluster, 100×100, 20% contiguous obstacle blocks, outer edge ring free). Real occlusion, preserved multi-path routing.

Figure 16. LOS occlusion on a cluster map: same agent FOV (r = 5) with (right) and without (left) the wall-blocking rule. Contiguous blocks occlude the supercover line where i.i.d. single cells do not.

E. Clusters at 3 UAVs (pass-2): the finding is swarm-size robust. The pass-2 rule ('A3 if the test is positive') triggers an A3 (3-UAV) replica on the same cluster-20% topology. RA's edge reproduces: RA vs FB **+2.6% (p = 0.0076, Holm-sig)**, RA vs CU **+6.3% (p = 0.0001, Holm-sig, the strongest delta of all cluster tests)**, no coverage regression in any cell. CU is again neutral (−1.2%, ns), consistent with the paper's dense regimes. The A6 cluster finding is not a 6-UAV artifact.

Cross-campaign verdict. The five-campaign dossier maps the boundary of the paper's claims: perfect maze = negative for both candidates (single-path routing collapse, not precision loss); 10% loops = neutral (penalty gone, no gain); H=16 = inert (horizon is not the lever); clusters 20% A6 = RA advantage intact (+3.0% Holm-sig, RA ≥ CU confirmed); clusters 20% A3 = RA advantage reproduced (+2.6% Holm-sig). The mechanistic story is now coherent across the whole corpus: open space and clusters (multi-path) keep the advantage, a perfect maze (single path) flips it negative, and loops (partial routing) sit between. What matters is routing redundancy, not spatial fragmentation per se. This is a post-submission confirmatory extension and is labeled as such; the paper's headline claims (sparse open space, Sections 6.4–6.10) are unchanged.

7. Discussion

The falsifications delimit where the signal works. Richness configurations fail as a direct target-selection rule (E1), as a mode-switching rule (E2/E3), and fail to move the saturated binary quality fraction (E4 primary). Each failure is informative: the bounded-horizon geometric frame already captures the coverage gains that these signals were expected to add, replicating the finding of our internal validation campaign. This is the “Occam’s razor” narrative made quantitative, and it is the honest baseline against which any positive claim must be read.

The positive effect is a budget effect, not an accuracy effect. In unbounded episodes every method reaches quality_final = 1.0 and residual bounds at parity. Under a fixed mission time, Coverage-U reduces the residual oracle bound by ~21% median with no coverage cost. The mechanism is visible in the traces: standard coverage spends remaining time reaching cheap frontier cells, leaving angular gaps; Coverage-U spends the same steps re-observing under-determined cells. The result matches the classical GDOP intuition [Bishop2004, Bishop2009] — accuracy is bought with baseline geometry — and shows that a *local, cheap* proxy can capture part of that intuition without a centralized planner.

Why the primary metric failed and what it teaches. quality_auc is a thresholded binary fraction that saturates; it cannot rank improvements below the well-localization threshold. The continuous residual bound is the metric aligned with the actual objective (accuracy). We report the preregistered primary as FAIL and the secondary as a confirmed discovery — a protocol lesson worth publishing: binary saturation metrics conceal accuracy effects that continuous bounds reveal.

Boundary conditions. The Coverage-U advantage is not statistically significant at 20% obstacle density (A6_obs020: +5.4%, ns after Holm correction; undetermined slightly reversed), marking a boundary of the effect: dense obstacles fragment the known-free space and dilute the signal. The mechanism is an integral-image dilution: the Coverage-U bonus scores under-observed cells through a fixed-area FOV window, so under dense obstacles the window's footprint is dominated by blocked cells — the `under_count_FOV` integral image is starved, the FOV area stays constant, and the average bonus per free cell collapses even though genuinely under-determined cells exist. Richness-Angular avoids this dilution because it scores the raw per-cell richness without averaging over the FOV area, which is why the angular-selection component keeps a significant +24% gain in the same dense regime (Section 6.6) while the plain under-coverage component saturates. The conclusion is therefore conditional: config-count prioritization helps in open, low-obstacle environments under time pressure — precisely the mission profile where partial coverage is unavoidable and residual accuracy matters most — and its angular variant extends the benefit to denser obstacle fields.

Topology: routing redundancy, not occlusion, is the boundary variable. The post-submission topology dossier (Section 6.11) isolates the mechanism. Under single-path routing (perfect maze), both candidates' open-space bonus flips sign — a significant coverage regression (not a precision one) driven by corridor over-lingering; re-opening 10% of the walls neutralizes the penalty without restoring any edge; a longer horizon is inert. When the same LOS occlusion is applied to contiguous blocks that preserve multi-path routing (cluster topology), Coverage-U returns to neutral and Richness-Angular's advantage is intact at 6 UAVs (+3.0%, Holm-sig) and reproduced at 3 UAVs (+2.6%, Holm-sig) with the directional RA ≥ CU comparison confirmed in both. The reading is that the under-observed/richness signals over-fit the open-space landscape and misjudge dead ends in single-path corridors, while their angular/richness content survives real occlusion when routing choice remains — refining the paper's boundary from 'spatial fragmentation' to 'routing redundancy'.

The dense-regime failure is a dilution, not a defect of the under-set signal. Because the dilution is a denominator artifact of the score (Section 4.4.1), it can be attacked directly: normalizing by the number of traversable cells in the footprint instead of the constant FOV area re-scales the bonus by what the sensor can actually observe, restoring signal strength in fragmented windows. We deliberately do not report results for that variant here — changing the score is a new intervention and deserves its own preregistered evaluation rather than a post-hoc rerun — but it is the first and cheapest fix on the path to dense-terrain operation, and the hybrid that switches to the angular signal when the local window is mostly blocked (Section 9) is the second.

8. Scope and boundaries

The paper answers a deliberately narrow, preregistered question: whether a config-count richness signal can reduce residual bearing-only localization error at equal coverage under a finite mission budget. Every boundary below is a scoping choice that keeps that question answerable with paired, preregistered evidence, and the robustness sweeps in Section 6.10 already relax the two idealizations that matter for the claim. What is not modeled is outside the claim, not a gap in it: within this framework the campaign is complete — falsifications, confirmation, robustness, and the local-vs-global ladder are all reported under the same locked protocol.

- Sensing is idealized where it is not the object of study. The decision signal is deliberately tested under self-localization noise (σ = 0.5, 1.0, Section 6.10), while the CRLB metric intentionally uses the true geometry so it measures localization work rather than sensor-model fidelity. Bearing/measurement noise and dropout sit outside the paper's question and would affect every method under comparison equally in the paired design.
- The environment is a single-scale grid with randomly placed obstacles — the topology the effect is claimed for. A post-submission topology dossier (Section 6.11) extends this with maze and cluster layouts under the same LOS model: the advantage survives occlusion when routing is preserved (clusters) and flips negative under single-path routing (perfect maze); multi-floor layouts and real terrain remain outside the claim.
- The confirmed effect is scoped to sparse-obstacle regimes under a finite mission budget — precisely the regime where coverage and accuracy compete. No claim of a general accuracy gain is made, and none is needed for the paper's conclusion.
- Communication is proximity-triggered fusion, the mechanism the decentralized claim rests on. Message topologies, delays, and bandwidth belong to the communication layer and are outside the finite-budget accuracy question.
- The centralized oracle (E5) assumes idealized perfect fusion; real centralized planners would face communication, latency, and map-fidelity costs not modeled here. The oracle's regression and the never-binding coverage guard are results reported and discussed in Sections 6.9 and 7, not limitations of this paper.
- Compute cost is measured per decision for the core comparison (Section 6.6): Coverage-U is at FB cost (2.17 ms vs 2.16 ms) and ~16% cheaper than the occupancy-entropy scorer (2.52 ms) on one serial benchmark. GDOP/FIM-style matrix inversions are not benchmarked directly; CU's advantage over them rests on its O(1) integral-image design plus the measured gap to the entropy scorer.

9. Future Work

E5 is complete: the centralized perfect oracle regresses both metrics, and the local-vs-global ladder (same config-count signal, +30% local vs regression global) locates the value in locality. E5-CORRECTED confirms the conclusion holds when the movement frame is held identical — the global under-set signal, not the oracle frame, is what collapses coverage. The coverage-guarded diagnostic did not bind in any real trajectory — a negative structural result that narrows the calibration-vs-structure question — so a definitive arbitration would need a guard that provably activates (e.g. inflated λ or a normed bonus). The immediate next step is therefore (i) a strict CPU-per-decision benchmark against GDOP/FIM planners; (ii) GPS-noise and communication-topology robustness (Phase 2 of the thesis plan) — Section 6.10 already extends self-localization noise and fusion range for the two confirmed regimes, and both robustness margins hold at moderate departures; (iii) the maze/cluster dossier (Section 6.11) shows the advantage survives occlusion when routing is preserved and fails under single-path routing — the immediate follow-up is multi-floor and room-and-door layouts to complete the routing-redundancy axis, plus the corridor-aware rescoring that addresses the dead-end misjudgment directly; (iv) adaptive λ (the Pareto plateau invites tuning, but we deliberately report the fixed preregistered value); (v) combining continuous prioritization with the deploy/orbit mechanics that failed as a pure mode switch — the positive result suggests the mechanics were sound and the gating was wrong; (vi) other richness estimators (ACE, Jackknife) and entropy baselines under the same budget protocol, to place the config-count proxy on the information-signal ladder.

Two follow-ups target the dense-regime boundary directly. First, the dynamic-normalization variant of Section 4.4.1 — normalize the Coverage-U bonus by traversable area instead of the constant FOV window — is the cheapest fix for the integral-image dilution diagnosed in Section 7, and deserves its own preregistered evaluation before any dense-terrain claim. Second, a low-cost hybrid policy that runs Coverage-U in open terrain and switches to the angular signal when the local window is mostly blocked: both methods share the same Frontier-Bounded movement frame, so the switch is a threshold on the local blocked fraction (e.g. free_count_FOV / FOV_area < 0.5, exactly the free-count statistic introduced in Section 4.4.1) rather than a new planner. Because Richness-Angular is the only method whose advantage survives 20% obstacle density and Coverage-U the stable, cheaper continuous weighting in open terrain (Section 10), the hybrid is the natural composition of the two positive results — a fallback, not a third method.

Finally, two steps move the evaluation off the synthetic grid. Real terrain: digital elevation models (MarsTrek, airborne LiDAR) would replace the random-obstacle maps with structured relief, testing the dense-regime claims under topography rather than uniform density. Real hardware: the measured per-decision cost of Coverage-U (2.17 ms, Section 6.6) fits the budget of an embedded flight controller, and a mini-drone fleet (e.g. Ryze Tello-class vehicles) or a Raspberry Pi 4B-class onboard processor is the concrete platform for a communication-limited outdoor trial. These are deliberately framed as validation steps — the claims of this paper are sim-based, and the paper says so (Section 8).

10. Conclusion

We asked whether a statistical-richness signal transposed to angular observation configurations can operate as a decision lever for multi-UAV bearing-only localization, evaluated honestly against a matched receding-horizon geometric control and a preregistered protocol. Three levers fail and one succeeds. Richness as target selection and as mode switching is falsified; continuous U-prioritized coverage (Coverage-U) reduces the residual oracle CRLB bound by a median 20.9% at equal coverage under finite mission budgets, robustly across λ, in sparse-obstacle environments. The centralized perfect oracle regresses both metrics, and the decisive comparison is the local-vs-global ladder: the same config-count signal gains ~+30% when scored locally and regresses when the same under-set is fused globally. E5-CORRECTED confirms the conclusion with the movement frame held identical: even in Coverage-U's own frame the global under-set signal regresses both metrics, so the effect is the signal's locality, not the oracle's frame. **Locality** is therefore the property that makes the trade-off attainable, not a limitation of the decentralized setting. The practical reading is sharp: when mission time is the scarce resource, spend the remaining coverage on cells that are angularly under-determined — and a cheap local singleton/doubleton count is sufficient to decide where. The two positive signals should be read as a trade-off rather than competing winners: Richness-Angular delivers the highest peak accuracy and is the most robust accuracy driver under spatial fragmentation — the only method whose edge survives 20% obstacle density — but as a raw target-selection signal it fails the preregistered primary and is prone to oscillation; Coverage-U provides a stable, equally cheap continuous weighting that is optimal in sparse environments and robust across its plateau. The universal takeaway is not which signal wins, but the **locality** of the decision signal: the same under-set gains ~+30% scored locally and regresses fused globally. The post-submission topology dossier (Section 6.11) sharpens the boundary: the advantage survives line-of-sight occlusion when routing choice is preserved (cluster topologies, reproduced at 3 and 6 UAVs) and flips negative only under single-path routing (perfect maze) — routing redundancy, not spatial fragmentation per se, is the condition for the trade-off. The headline claims of the paper (sparse open space, Sections 6.4–6.10) are unchanged.

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